

# ODRE PQC v0.2.9

## Product Overview & Security Architecture

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Turn complex post-quantum security into a simple application security boundary.

**ODRE handles the PQC complexity. Developers install it and connect the application.**

## What ODRE PQC is

ODRE PQC is not merely a library of post-quantum algorithms. It is designed as a Cross-Platform Application Security Boundary placed in front of protected application routes.

## Easy installation, integration and operation

The product hides the complexity of native cryptographic runtimes, key and session lifecycle, replay and sequence enforcement, filesystem trust, operational status and restart behavior behind a compact operator and developer experience.

```
pip install → odre-pqc init → odre-pqc doctor → odre-pqc verify
```

```
from odre_pqc import install | install(app)
```

## Cross-Platform

ODRE PQC v0.2.9 is verified on Windows Server 2022 AMD64 and Ubuntu 24.04 LTS AMD64. OS-specific DLL/ACL/reparse behavior and Linux ELF/symlink/inode/FD/POSIX behavior are handled inside platform security layers.

## Real PQC Runtime

The verified runtime uses OpenSSL 3.5.8 with ML-KEM-768 and ML-DSA-65. Real self-tests and hybrid handshakes passed on both supported operating systems, with no automatic classical-only downgrade.

## Fail-Closed Security Boundary

If protection cannot be proven healthy, protected execution is blocked. Runtime identity, filesystem identity, session, sequence and authenticated status are treated as security conditions rather than best-effort signals.

## Beyond body encryption

The security boundary covers application semantics, not only an encrypted HTTP body. Security-relevant request semantics and response metadata such as protected headers and cookies are bound to the protected contract.

## Replay, Tamper, Sequence and Session Defense

The final development audit exercised replay concurrency, strict sequence, tamper, route/session substitution and semantic mutation. Windows processed 1,280 duplicate requests and Ubuntu 2,880, with zero double handler execution.

## Native Runtime Trust

Windows native stress performed 3,000 actual provider/WinDLL load attempts with zero untrusted DLL load, untrusted dependency load, reparse bypass or verified-vs-loaded identity mismatch. Linux native trust testing covered file/parent/symlink swaps and dependency/provider injection.

## Crash, SIGKILL and Restart Security

Lifecycle security includes graceful shutdown, forced termination, SIGKILL and restart. The design prevents stale sessions, stale worker leases, false stale status and security-state corruption from being treated as healthy protection.

## Release-Clean Binary

Shipping Windows DLLs are release-clean. The verified v0.2.9 artifact contains zero debug sections, DWARF markers, debug payload, absolute build/source paths and private build-topology leakage.

## Hardened through adversarial testing

ODRE PQC was repeatedly tested with static review and live adversarial execution including replay races, TOCTOU, native replacement, dependency hijacking, status spoofing, fuzzing, HTTP smuggling and crash recovery. Confirmed defects were remediated while failed artifact identities and evidence were preserved.

## v0.2.9 Verification Summary

The v0.2.9 full development security audit completed with Windows PASS, Ubuntu PASS, and zero new unresolved Critical, High or Medium product findings.

## Exact Artifact Identity

The exact verified wheel is identified by SHA-256:

A8AFA10EE0AFACEF1C0FAF55FF07B96C722920E5AEE8692F6F026E804F028697. Audit start and end identities matched.

## What ODRE PQC is designed to deliver

ODRE PQC is intended to make deployable post-quantum protection practical for application developers without requiring them to assemble the cryptographic and operating-system security machinery themselves.